Vision Transformer

안동주

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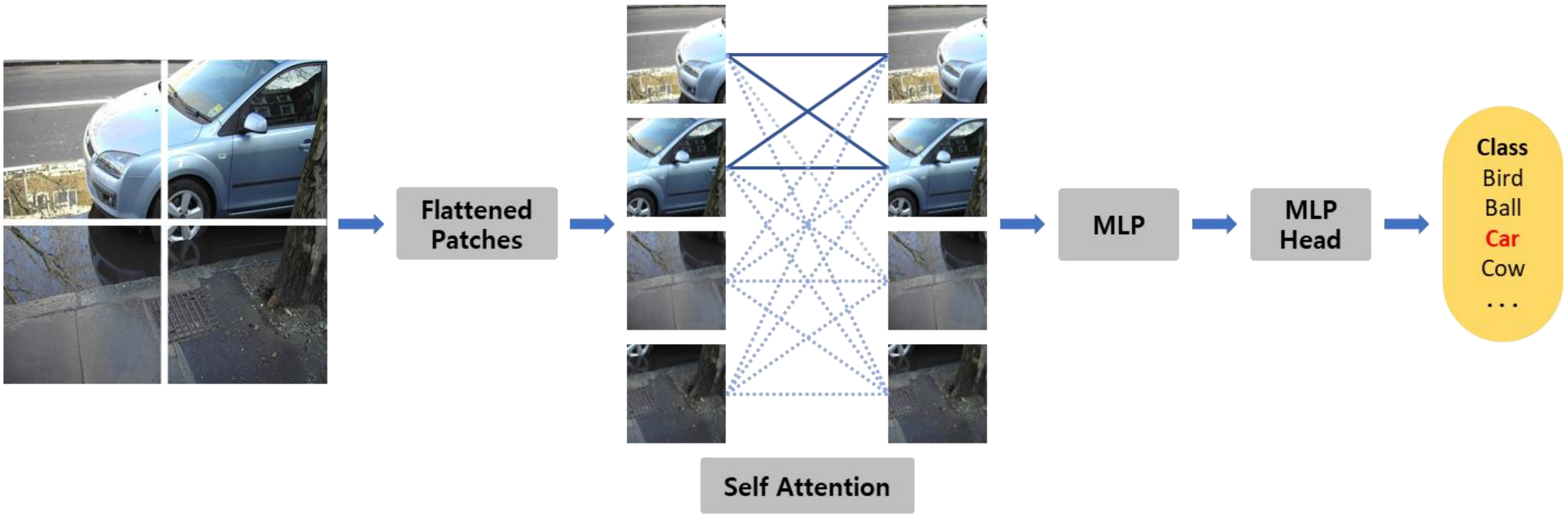
Transformer　ViT　실험 결과

Introduction

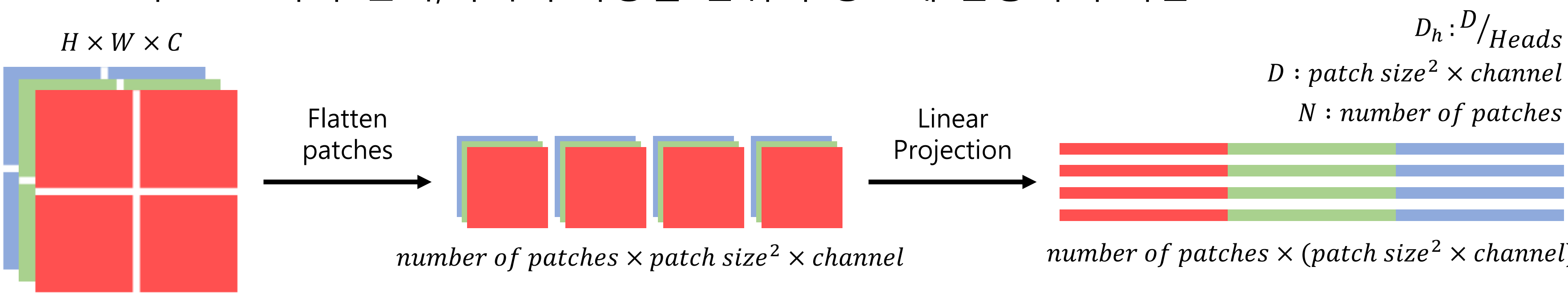
- **Transformer**
  - Sequence를 patch로 나누고, 각 patch의 연관도를 계산
  - Self Attention으로 가장 낮은 층에서도 전체 이미지를 통합하여 학습 가능

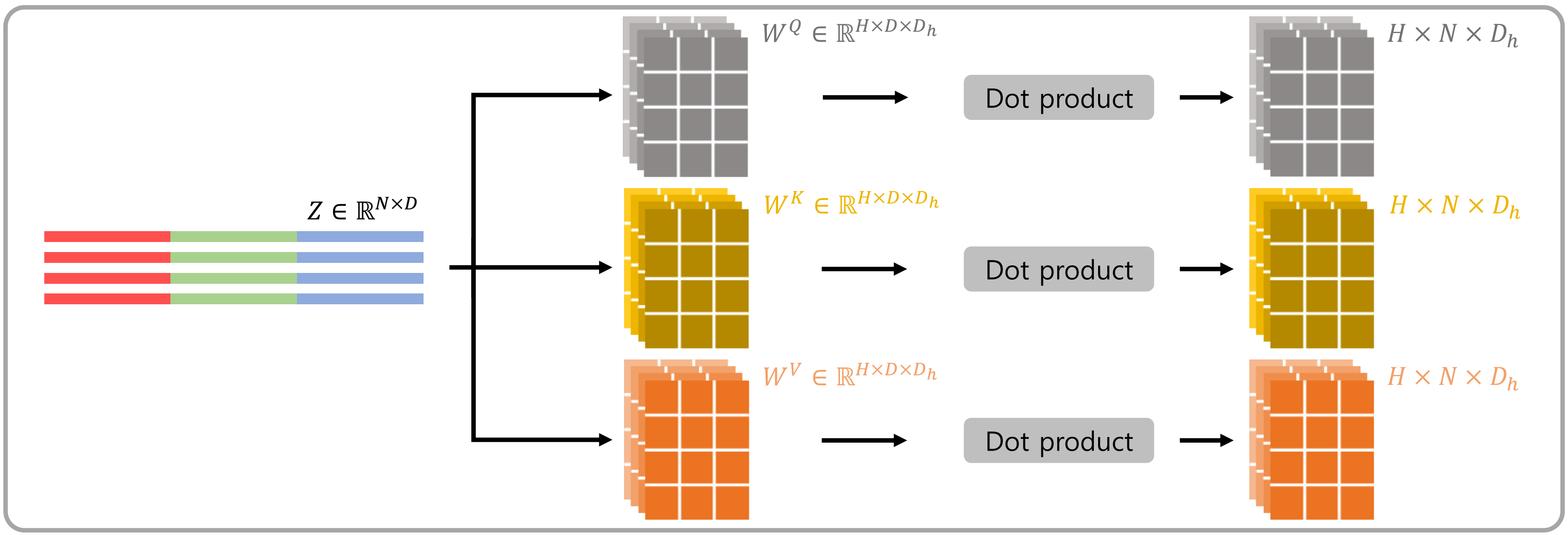


Transformer

▪ Multi-Head Attention

- 각 역할에 맞는 Query, Key, Value를 입력 토큰을 사용해 선형 변환
- 각 Head마다 전역,지역의 다양한 범위의 정보에 집중하여 학습

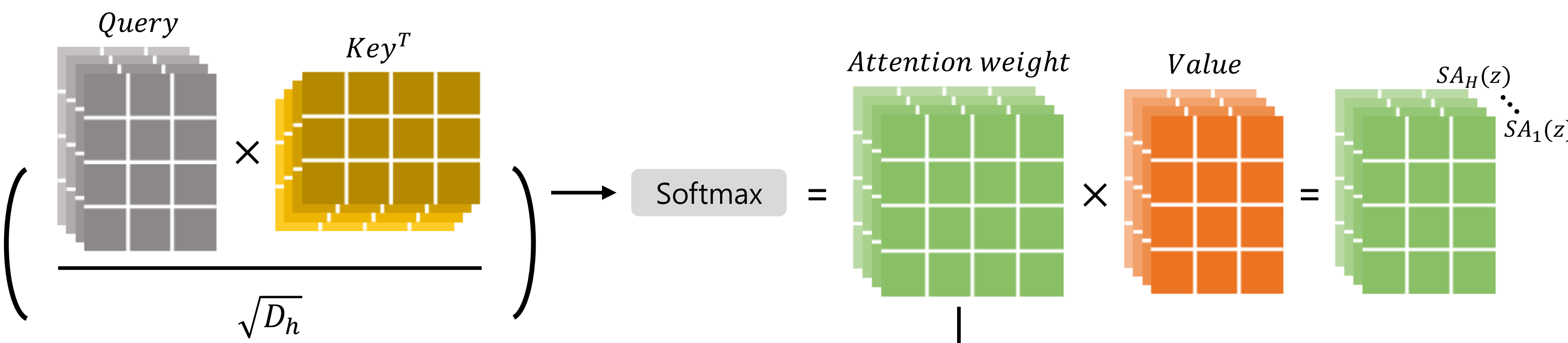


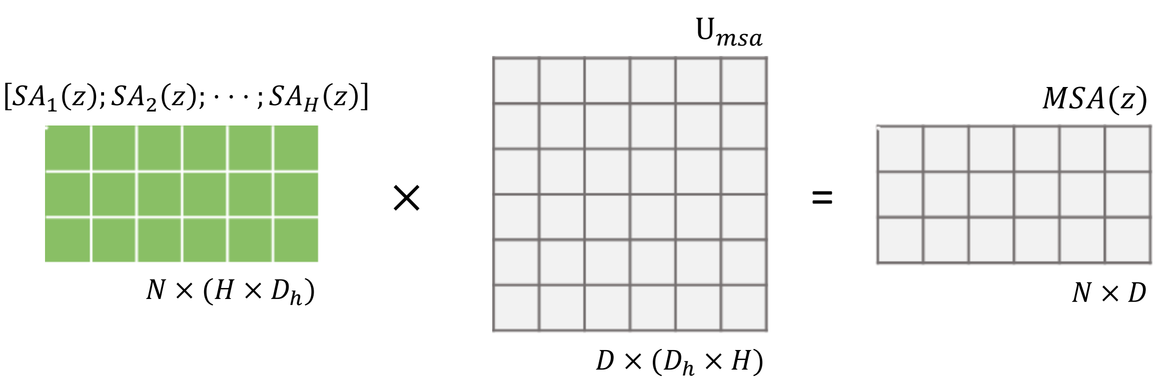


Transformer

▪ Multi-Head Attention

- 기울기 소실을 막기 위해 $\sqrt{D_h}$로 scaling
- Attention weight는 query와 key의 pairwise similarity에 기반



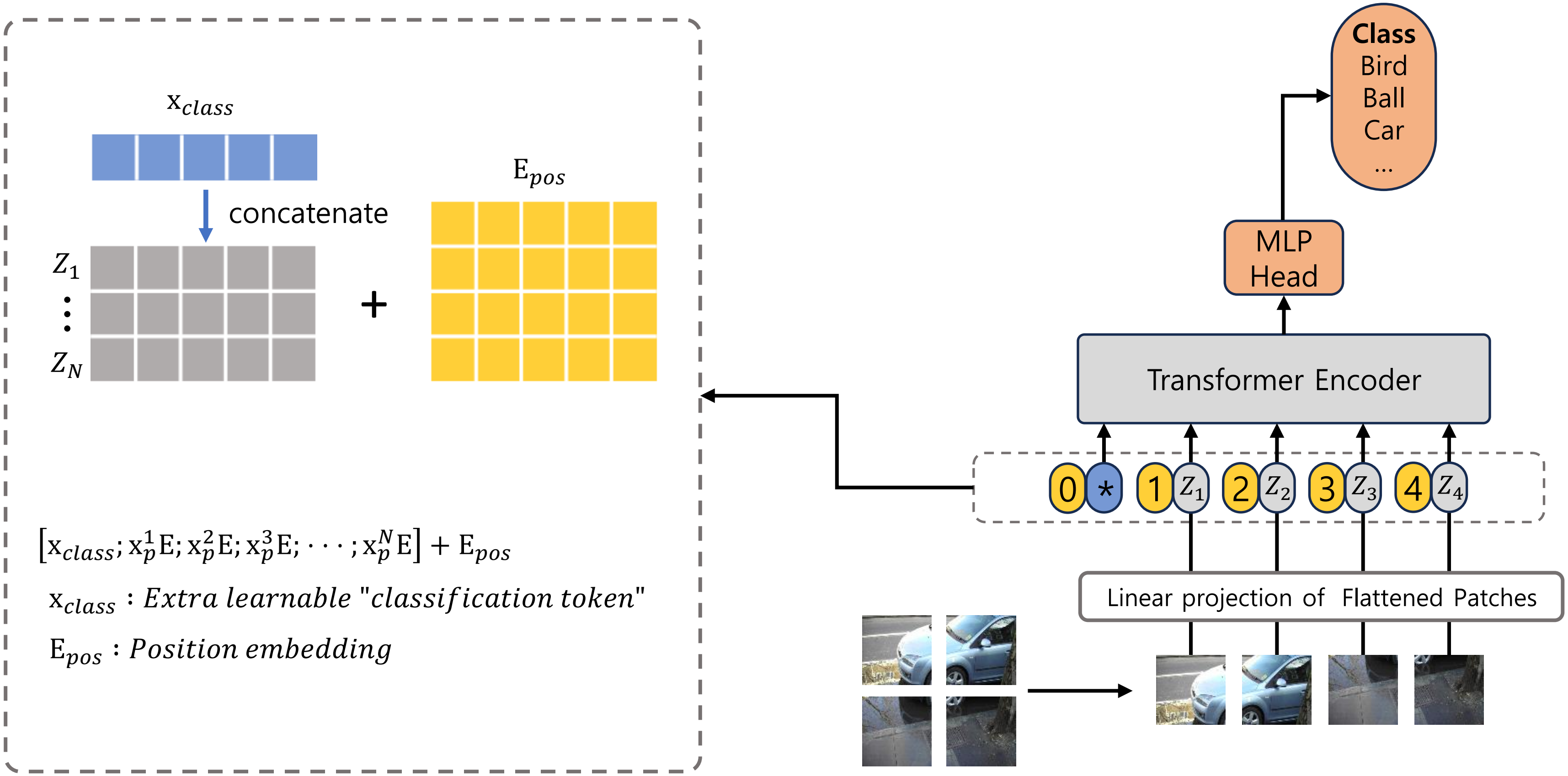


$$[SA_1(z); SA_2(z); \cdots; SA_1(z)]\mathrm{U}_{msa} = MSA(z)$$

	K_{p1}	K_{p2}	K_{p3}	K_{p4}
Q_{p1}	0.18	0.52	0.15	0.15
Q_{p2}	0.44	0.22	0.19	0.15
Q_{p3}	0.13	0.18	0.12	0.57
Q_{p4}	0.21	0.14	0.49	0.16

Introduction

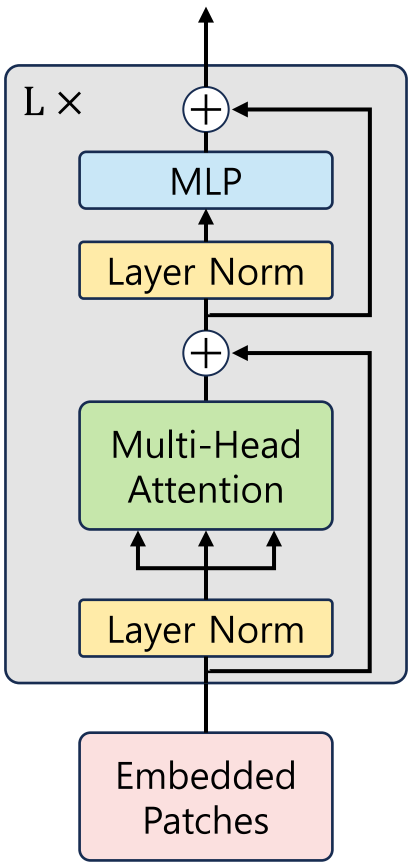
- **ViT[1]**
  - 학습 가능한 embedding을 사용해 최종 classification 수행
  - 위치 정보 유지를 위해 position embedding을 더함

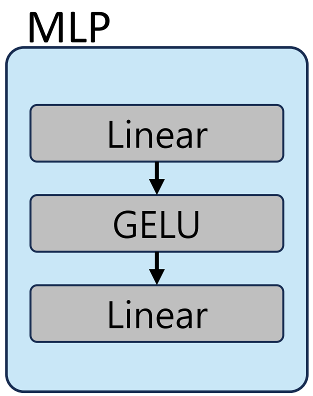


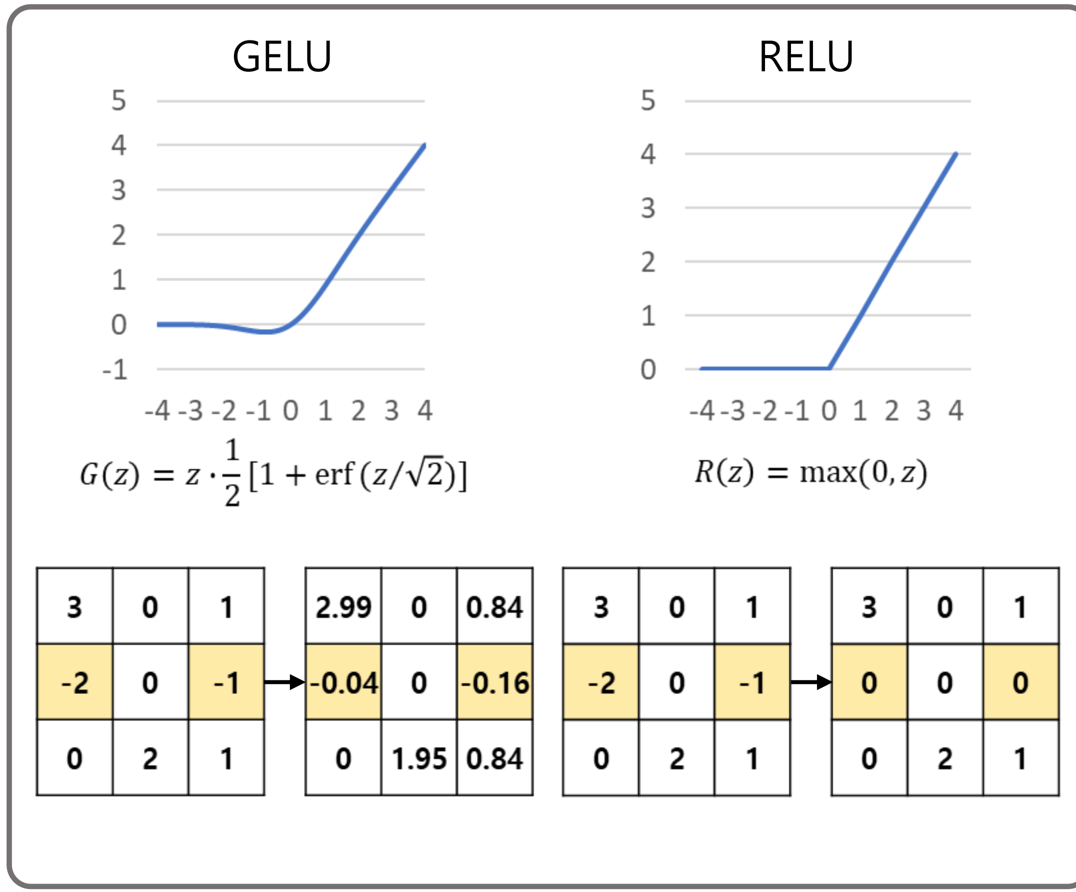
[1] A. Dosovitskiy, L. Beyer et al. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." In *International Conference on Learning Representations*. 2020

ViT

- **Activation function**
  - RELU는 음수 입력에서 0이 되어 gradient가 0이 되어 weight 업데이트 안됨
  - 음수 값 보존을 위해 GELU 사용



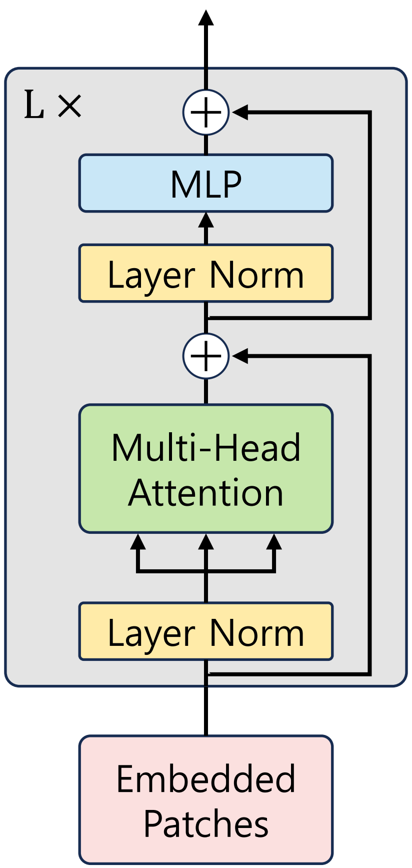




ViT

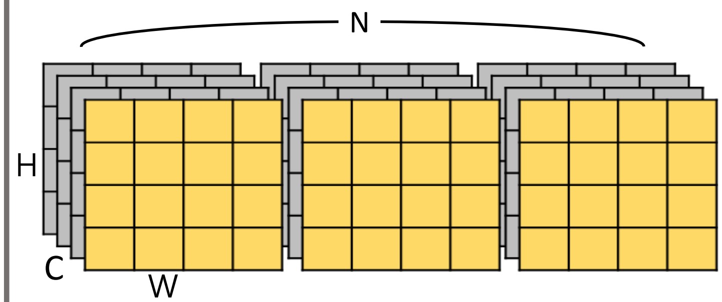
▪ Normalization

- Layer normalization은 batch 크기에 무관하게 정규화



Batch normalization

$Batch = 3 \quad m = N \times H \times W$

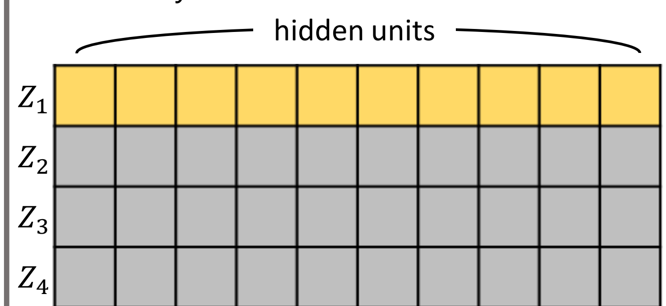


$$\mu_B = \frac{1}{m}\sum_{i=1}^{m} x_i$$

$$\sigma_B^2 = \frac{1}{m}\sum_{i=1}^{m} (x_i - \mu)^2$$

Layer normalization

$hidden\ units = P^2 \times C$



$$\mu_l = \frac{1}{H}\sum_{i=1}^{H} x_i$$

$$\sigma_l^2 = \frac{1}{H}\sum_{i=1}^{H} (x_i - \mu_l)^2$$

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \varepsilon}}$$

$$y_i = \gamma\, \hat{x}_i + \beta$$

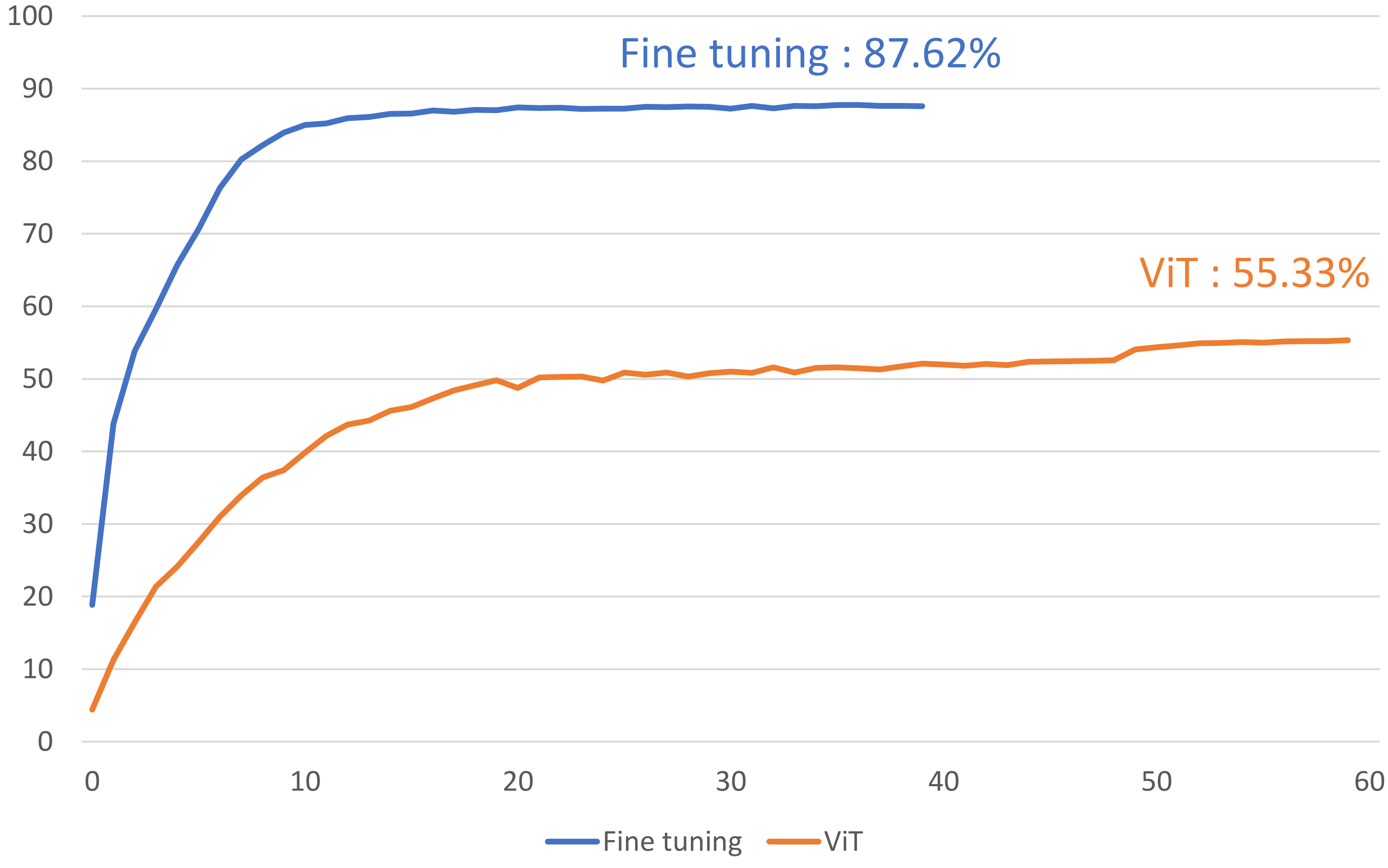
Experiment

- **Setting**

	ViT	**Fine-tuning ViT**
Dataset	TinyimageNet	TinyimageNet
Pre-trained	X	ImageNet1k
Epoch	60	40
Batch size	256	256
Learning rate	0.3	0.003
Learning rate scheduling	Multiply by 0.1 at 50 epochs	X
optimizer	SGD	SGD
Augmentation	Random horizontal Flip Random Crop	Random horizontal Flip Random Crop

Result

- **ViT & Fine tuning**
  - Fine tuning : 87.618%
  - ViT : 55.329%



The end

Thank you

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